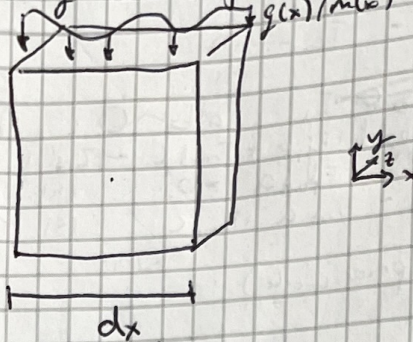


TITLE Derivation of beam bending PROJECT

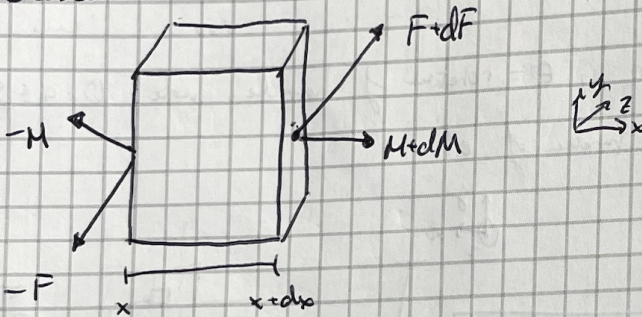
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Take an arbitrary beam subject to an external force & external moment



$g(x)$ is the external ^{force} vector
 $m(x)$ is the external moment vector

Due to these applied loads we see internal forces & moments in this element



These force and moment vectors can be represented as vectors in 3-dimensional space as

$$\vec{F} = \vec{F}_{shear} = (F_x, F_y, F_z) \quad \vec{F}(x) = (F_x, F_y, F_z)$$

We assume in this case that loading only varies along y - for now.

$$\vec{M} = M = (M_x, M_y, M_z)$$

↑
torsional moment

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We will assume that deflections remain small $\Rightarrow$ linear

$$\sum \vec{F} = 0$$

So we take at other as see $\sum \vec{F} = m\vec{a} = 0 = -F + F + dF - gdx = 0$.

$$\Rightarrow d\vec{F}/dx + \vec{g} = 0$$

$$\& \underbrace{-M + M + dx + i dx \times (F + dF)}_{\text{applies moment}} + \underbrace{m dx}_{\text{moment from force vector}} = 0$$

$$\Rightarrow \frac{\partial M}{\partial x} + i \times F + m = 0$$

$$i \times F = i \times (N\hat{i} + Q_y\hat{j} + Q_z\hat{k}) = -Q_z\hat{j} + Q_y\hat{k}$$

$$\Rightarrow \partial F_x / \partial x + g_x = 0, \partial F_y / \partial x + g_y = 0, \partial F_z / \partial x + g_z = 0$$

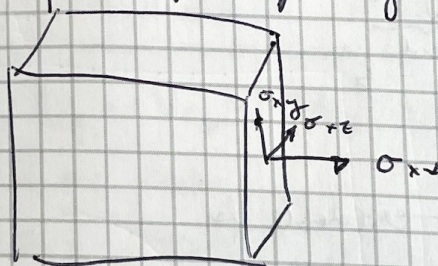
$$\& \partial M_x / \partial x + m_x = 0, \partial M_z / \partial x + Q_y + m_z = 0, \partial M_y / \partial x - Q_z + m_y = 0$$

Now using this we can figure out the corresponding stress in the beam.

$$F = \int \sigma dA$$

$$\Rightarrow F_x = \int \sigma_{xx} dA, F_y = \int \sigma_{xy} dA, F_z = \int \sigma_{xz} dA$$

$$M_y = \int z \sigma_{xx} dA, M_z = - \int y \sigma_{xx} dA$$



$$M_y = \int z \sigma_{xx} dA \& M_z = - \int y \sigma_{xx} dA$$

$$\vec{\sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$

$\sim$ symmetric +ve definite matrix

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Assume cross section orthogonal to y axis for our CRF

Now for our CRF

assume the cross section is orthogonal to x -axis. The xy -coordinates remain plane & keeps its shape during deformation

$\Rightarrow$ No poisson contraction

$\Rightarrow$ The final cross-section is defined by a translation & rotation vector

$$w = \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} \quad \& \quad \Theta = \begin{bmatrix} \Theta_x \\ \Theta_y \\ \Theta_z \end{bmatrix}$$

Note $\|w\|_2 \ll l$ (length of beam where l is

$\Rightarrow \sin \Theta \approx \tan \Theta \approx \Theta$ for future calculations

Now we let

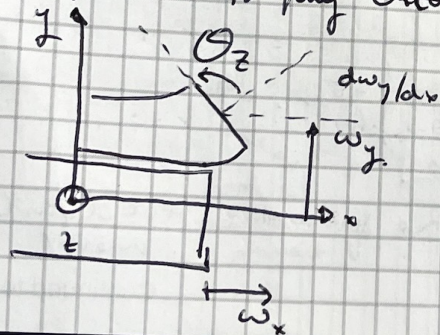
$\vec{u}(x, y, z) = (u_x, u_y, u_z)$ be the displacement vector defining the change in position of the reference face

$$\Rightarrow u_x = \underbrace{w_x}_{\text{axial bending}} + \underbrace{z \Theta_y}_{\text{rotation}} + \underbrace{y \Theta_z}_{\text{bending in } z}$$

$$u_y = w_y - z \Theta_x$$

$$u_z = w_z + y \Theta_x$$

$\Rightarrow$ deformation is fully calculable for



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We can now figure out the corresponding strains.

$$\begin{aligned}\epsilon_{xx} &= \frac{\partial u_x}{\partial x} = \frac{\partial u_x}{\partial x} + z \frac{\partial \theta_y}{\partial x} - y \frac{\partial \theta_z}{\partial x} \\ \gamma_{xy} &= \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} = \frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y} + z \frac{\partial \theta_z}{\partial x} - \frac{\partial \theta_y}{\partial x} \\ \gamma_{xz} &= \frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} = \frac{\partial u_z}{\partial x} + \frac{\partial \theta_z}{\partial x} y + \theta_y + \frac{\partial u_z}{\partial x}\end{aligned}$$

$$\Rightarrow \gamma_{xy}(x, z) \text{ \& } \frac{\partial \sigma}{\partial y} = 0.$$

$\Rightarrow$ As γ_{xy} is not a function of $y \Rightarrow$ shear stress is constant over cross section.

However for non-circular cross-sections we have

$$u_x(x, y, z) = w_x(x) + z \theta_y(x) - y \theta_z(x) + \underbrace{w(y, z)}_{\text{warping function}} \frac{\partial \theta_z}{\partial x}$$

$$\Rightarrow \epsilon_{xx} = \partial_x u_x = \partial_x w_x + z \partial_x \theta_y - y \partial_x \theta_z + \partial_x w \frac{\partial \theta_z}{\partial x}$$

$$\gamma_{xy} = \partial_y u_x + \partial_x u_y = \partial_x w_y - \theta_z - (\partial_y w - z) \partial_x \theta_x$$

$$\gamma_{xz} = \partial_z u_x + \partial_x u_z = \partial_x w_z + \theta_y + (\partial_z w + y) \partial_x \theta_x$$

Now the radius of curvature of the beam. (assume) $\approx$

$$\underbrace{r_y d\theta_z = dx \text{ \& } r_z d\theta_y = dx.}$$

$$K_y = -1/r_z = d\theta_z/dx \text{ \& } K_z = 1/r_y = d\theta_y/dx.$$

Assume we are dealing with Euler-Bernoulli. (x-direction free & small)

$$\Rightarrow \theta_y = -dw_z/dx \text{ \& } \theta_z = dw_y/dx.$$

This is valid as F_x very small. $\Rightarrow$ face is $\approx$ normal to beam.

$$\Rightarrow \epsilon_{xx} = \underbrace{\epsilon(x)}_{\partial w_x / \partial x} + z \underbrace{K_y}_{\partial \theta_y / \partial x} - y \underbrace{K_z}_{\partial \theta_z / \partial x} + \underbrace{w(y, z)}_{\text{curvature terms}} \partial_x \theta_x.$$

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$$F_x = \int \sigma_{xx} dA = \int E \left[\frac{\partial u}{\partial x} + z \frac{\partial \theta_y}{\partial x} - y \frac{\partial \theta_z}{\partial x} + w(y, x) \frac{\partial \theta_x}{\partial x} \right] dA.$$

$$= E \left[\frac{\partial u}{\partial x} A + \underbrace{S_y \frac{\partial \theta_y}{\partial x}}_{\text{shear stress}} - \underbrace{S_z \frac{\partial \theta_z}{\partial x}}_{\text{shear stress}} - \underbrace{S_w \frac{\partial \theta_x}{\partial x}}_{\text{shear stress}} \right].$$

$$F_y = \int \sigma_{xy} dA = \int G \left(\frac{\partial u}{\partial y} - \theta_z + (\frac{\partial w}{\partial y} - z) \frac{\partial \theta_x}{\partial y} \right) dA$$

$$= G \left(A_y \left(\frac{\partial u}{\partial y} - \theta_z \right) + \underbrace{R_y}_{\text{shear stress}} \frac{\partial \theta_x}{\partial y} \right)$$

$$R_y = \int (\frac{\partial w}{\partial y} - z) dA$$

$$F_z = G \left(A_z \left(\frac{\partial u}{\partial z} + \theta_y \right) + R_z \frac{\partial \theta_x}{\partial z} \right).$$

$$M_x = \int \sigma_{xx} y - \sigma_{xy} z dA = \int G \left(y \left(\frac{\partial u}{\partial y} - \theta_z + (\frac{\partial w}{\partial y} - z) \frac{\partial \theta_x}{\partial y} \right) - z \left(\frac{\partial u}{\partial z} + \theta_y + (\frac{\partial w}{\partial z} - y) \frac{\partial \theta_x}{\partial z} \right) \right) dA.$$

$$= G \left(S_z \left(\frac{\partial u}{\partial z} + \theta_y \right) - S_y \left(\frac{\partial u}{\partial y} - \theta_z \right) + \underbrace{K \frac{\partial \theta_x}{\partial x}}_{\text{torsional stiffness}} \right)$$

$$M_y = \int z \sigma_{xy} dA = \int E \left[\frac{\partial u}{\partial y} + z \frac{\partial \theta_y}{\partial y} - y \frac{\partial \theta_z}{\partial y} + w(y, x) \frac{\partial \theta_x}{\partial y} \right] dA$$

$$= E \left(S_y \frac{\partial u}{\partial y} + I_{yy} \frac{\partial \theta_y}{\partial y} - I_{yz} \frac{\partial \theta_z}{\partial y} + I_{wy} \frac{\partial \theta_x}{\partial y} \right)$$

$$M_z = E \left(-S_z \frac{\partial u}{\partial z} - I_{yz} \frac{\partial \theta_y}{\partial z} + I_{zz} \frac{\partial \theta_z}{\partial z} - I_{wz} \frac{\partial \theta_x}{\partial z} \right)$$

$$A_y = \alpha_y A \quad \& \quad A_z = \alpha_z A.$$

(shear area).

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These correction factors for A_y & A_z come from the fact that the shear stress distribution is not linear as we assumed, it is in fact parabolic.

⇒ We get

$$\begin{bmatrix} F_x \\ M_y \\ M_x \\ F_y \\ F_z \end{bmatrix}$$

$$= \begin{bmatrix} EA - E\gamma_y - E\gamma_z & 0 & 0 \\ E\gamma_y & EI_{yy} & -EI_{yz} \\ -E\gamma_z & -EI_{yz} & EI_{zz} \\ 0 & 0 & 0 & GA_y & 0 \\ 0 & 0 & 0 & 0 & GA_z \end{bmatrix} \begin{bmatrix} \frac{\partial w}{\partial x} \\ \frac{\partial \theta_y}{\partial x} \\ \frac{\partial \theta_z}{\partial x} \\ \frac{\partial w}{\partial x} - \theta_z \\ \frac{\partial w}{\partial x} + \theta_y \end{bmatrix}$$

we ignore θ_x

term is ignored

symmetric

Remember: $\frac{\partial F_x}{\partial x} = -f(x)$, $\frac{\partial F_y}{\partial x} = -\frac{f(y)}{x}$, $\frac{\partial F_z}{\partial x} = -\frac{f(z)}{x}$

$$\frac{\partial M_y}{\partial x} = F_z - m_y$$

$$\frac{\partial M_z}{\partial x} = -F_y - m_z$$

we get

$$\begin{bmatrix} 0 \\ F_z \\ -F_y \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} f(x) \\ m_y \\ m_z \\ \frac{\partial F_y}{\partial x} \\ \frac{\partial F_z}{\partial x} \end{bmatrix} = \frac{\partial}{\partial x} \begin{bmatrix} F_x \\ M_y \\ M_z \\ \frac{\partial w}{\partial x} \\ \frac{\partial w}{\partial x} \end{bmatrix} = \frac{\partial}{\partial x} (A B)$$

moment & force distributions

$$\Rightarrow \frac{\partial}{\partial x} \begin{bmatrix} EA \frac{\partial w}{\partial x} + E\gamma_y \frac{\partial \theta_y}{\partial x} - E\gamma_z \frac{\partial \theta_z}{\partial x} \\ E\gamma_y \frac{\partial w}{\partial x} + EI_{yy} \frac{\partial \theta_y}{\partial x} - EI_{yz} \frac{\partial \theta_z}{\partial x} \\ -E\gamma_z \frac{\partial w}{\partial x} - EI_{yz} \frac{\partial \theta_y}{\partial x} + EI_{zz} \frac{\partial \theta_z}{\partial x} \\ GA_y (\frac{\partial w}{\partial x} - \theta_z) \\ GA_z (\frac{\partial w}{\partial x} + \theta_y) \end{bmatrix}$$

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$$= \left[\begin{aligned} & \epsilon A \frac{\partial^2 w_x}{\partial x^2} + \epsilon S \left(\frac{\partial^2 \theta_y}{\partial x^2} - \epsilon S_2 \frac{\partial^2 \theta_z}{\partial x^2} \right) \\ & G A_y \left(\frac{\partial^2 w_x}{\partial x^2} - \frac{\partial \theta_z}{\partial x} \right) \\ & G A_z \left(\frac{\partial^2 w_x}{\partial x^2} - \frac{\partial \theta_y}{\partial x} \right) \end{aligned} \right]$$

$$= \left[\begin{aligned} & 0 \\ & 0 \\ & 0 \\ & -G A_y \frac{\partial \theta_z}{\partial x} \\ & -G A_z \frac{\partial \theta_y}{\partial x} \end{aligned} \right]$$

let $\begin{bmatrix} 0 \\ F_x \\ -F_y \\ 0 \end{bmatrix} - \begin{bmatrix} F_x \\ w_x \\ w_y \\ w_z \end{bmatrix} = \vec{S} - \vec{f} = \vec{S}/\partial x$

where $\vec{S} = \begin{bmatrix} \partial F_x / \partial x \\ \partial w_x / \partial x \\ \vdots \end{bmatrix}$

note $\vec{S} = \frac{d}{dx} \vec{AB} \Rightarrow \vec{B} = \vec{E} = \begin{bmatrix} \partial w_x \\ \partial \theta_y \\ \partial \theta_z \end{bmatrix}$
 Constant wrt x

$\Rightarrow \frac{d\vec{S}}{dx} = \frac{d}{dx} \vec{AB} = A \frac{d}{dx} \vec{B} = \frac{d\vec{E}}{dx}$

$\Rightarrow \frac{d\vec{E}}{dx} = \begin{bmatrix} \partial_{xx} w_x \\ \partial_{xx} \theta_y \\ \partial_{xx} \theta_z \\ \partial_{xx} w_y - \partial_x \theta_z \\ \partial_{xx} w_z - \partial_x \theta_y \end{bmatrix}$

from before $F_y = -G A_y \epsilon_4$ & $F_z = G A_z \epsilon_5$

$\Rightarrow \vec{S} = \begin{bmatrix} 0 \\ G A_z \epsilon_5 \\ -G A_y \epsilon_4 \end{bmatrix} \Rightarrow \vec{S} = C \epsilon \Rightarrow C = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & G A_z \\ 0 & 0 & 0 & -G A_y & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\Rightarrow K \frac{d\epsilon}{dx} = C \epsilon - f$ as required.

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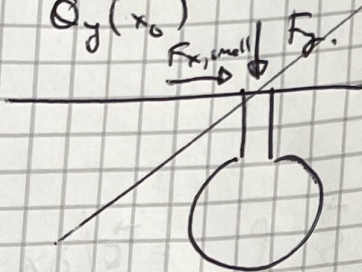
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Now we have defined the boundary conditions for this beam with our prior work.
we have a normal force, a shear force & one direction

$$F_x = F_{q, \text{cen}}$$

$$Q_y(x_0)$$



$$\Rightarrow \frac{\partial F_x}{\partial x} = -q_x h$$

$$\frac{\partial M_y}{\partial x} = GA_z \left(\frac{\partial u_y}{\partial x} + \theta_z \right)$$

$$\frac{\partial M_y}{\partial x} = GA_z \left(\frac{\partial u_y}{\partial x} + \theta_z \right) - m_y$$

$$\frac{\partial M_z}{\partial x} = GA_y \left(\frac{\partial u_z}{\partial x} - \theta_y \right) - m_z$$

$$\frac{\partial F_y}{\partial x} = -q_y - f_y$$

$$\frac{\partial F_z}{\partial x} = -f_z$$

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